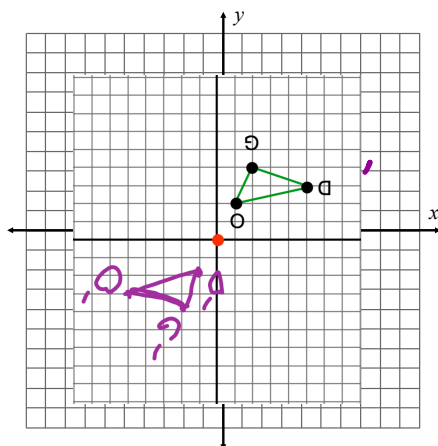


Students will describe transformations.

Do Now

1) Graph, label, and state the coordinates of $\triangle D'O'G'$, the image of $\triangle DOG$ after being rotated 180° about the origin.

$$R_{180} \rightarrow (x, y) \rightarrow (-x, -y)$$

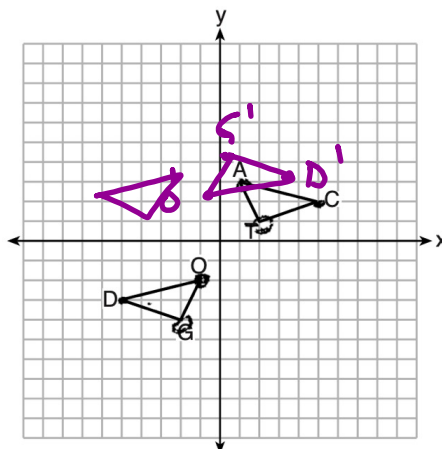


$$\begin{aligned} D(-5, -3) &\rightarrow D'(1, 2) \\ O(-1, -2) &\rightarrow O'(5, 3) \\ G(-2, -4) &\rightarrow G'(2, 4) \end{aligned}$$

If done early, graph $\triangle D''O''G''$, the image of $\triangle DOG$ after being rotated 270° about the origin.

2) Why did this student receive 0/2 points on this test question?

25 On the set of axes below, $\triangle DOG \cong \triangle CAT$.



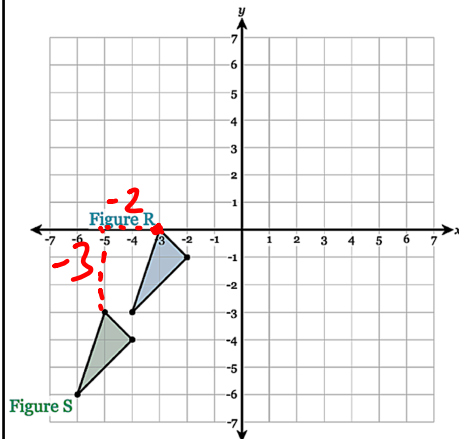
Describe a sequence of transformations that maps $\triangle DOG$ onto $\triangle CAT$.

$\triangle DOG$ is rotated 180° around the origin.

Describing Transformations

To describe a **translation**, you must have a vertical direction and distance and a horizontal direction and distance.

Ex 1: Describe a transformation that maps Figure R onto Figure S.

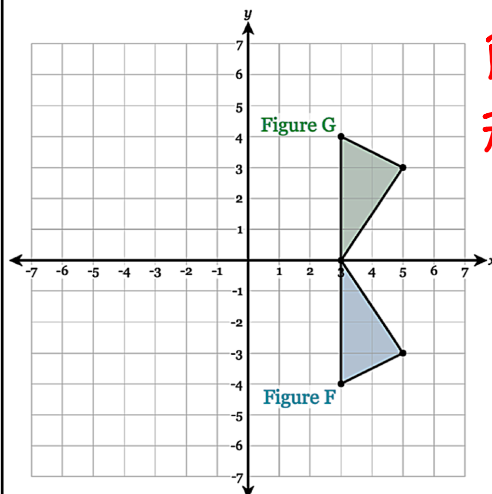


Shift Figure
R left 2
and down
3 units
to map onto
figure S.



To describe a **reflection**, you must provide a line
of reflection that you are reflecting over.

Ex 2: Describe a transformation that maps Figure F onto Figure G.



Reflect Figure
F across the
X-axis.

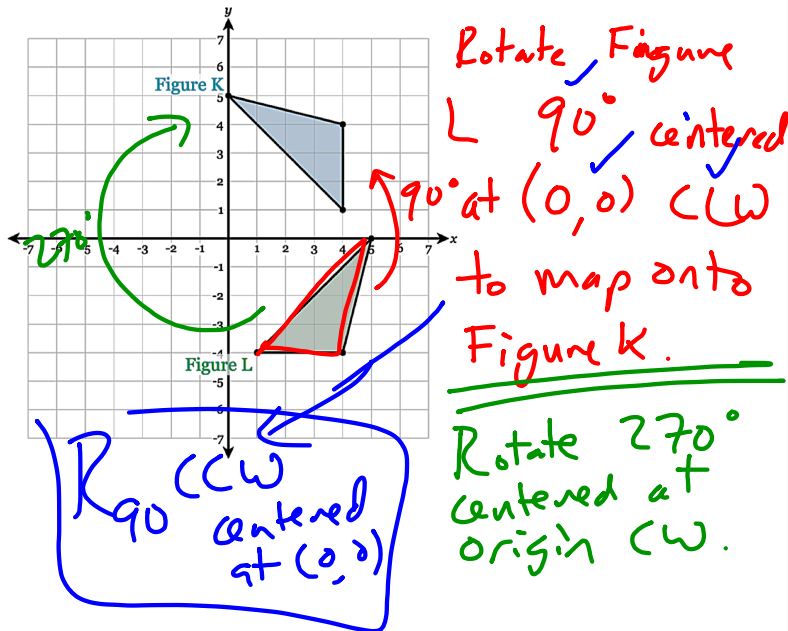
or

X-axis

To describe a **rotation**, you must provide three piece of information:

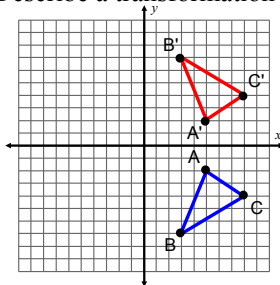
- 1) Center and 2) Direction and 3) How far.
 (origin) (CCW or CW) ($90^\circ, 180^\circ, 270^\circ$)

Ex 3: Describe a transformation that maps Figure L onto Figure K.



Practice

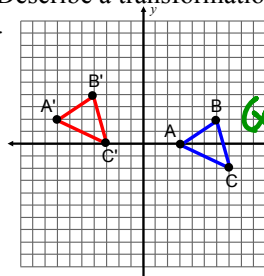
Ex 4: Describe a transformation that maps $\triangle ABC$ onto $\triangle A'B'C'$ in the graph below.



Reflected figure ABC across the x-axis

Preserves orientation? Y N

Ex 5: Describe a transformation that maps $\triangle ABC$ onto $\triangle A'B'C'$ in the graph below.



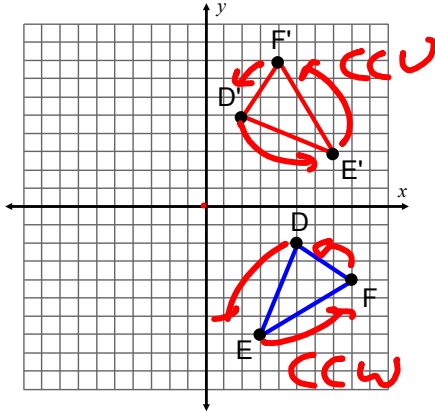
10 to the left and 2 up

$(x, y) \rightarrow (x-10, y+2)$

$\langle -10, 2 \rangle$

Preserves orientation? Y N

Ex 6: Describe a transformation that maps $\triangle DEF$ onto $\triangle D'E'F'$ in the graph below.



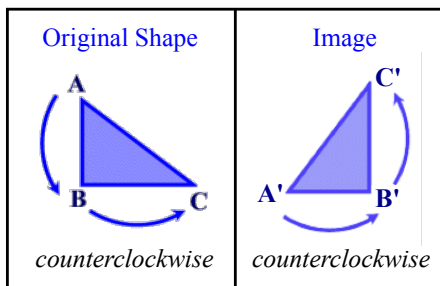
Rotate $\triangle DEF$
 90° CCW
 around the
 origin.

Preserves orientation? ☒ Y ☐ N

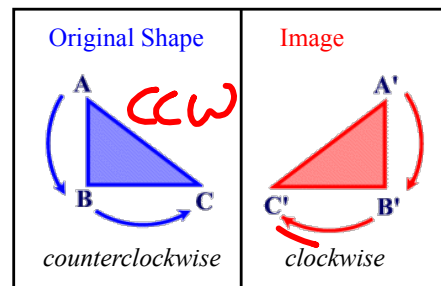
Orientation

A shape's **orientation** has to do with the order of its vertices

If the image of a shape after some transformation has the *same order* of *vertices* as the original shape, then we say that *orientation is preserved*.



Orientation
IS
 Preserved

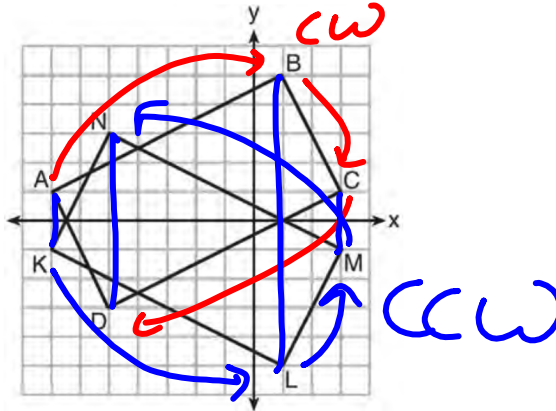


Orientation
NOT
 Preserved

Reflections do NOT preserve orientation!

Ex 7:

On the set of axes below, rectangle $ABCD$ can be proven congruent to rectangle $KLMN$ using which transformation?



- 1) rotation
- 2) translation
- 3) reflection over the x -axis
- 4) reflection over the y -axis

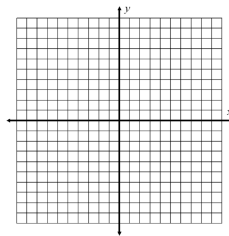
Geometry

Hr:

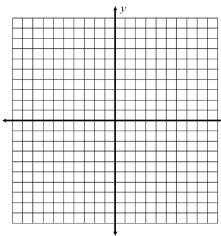
Name: _____

Assignment 2.03: Rotations

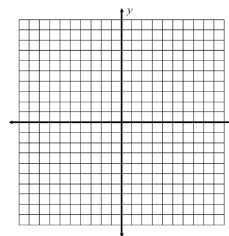
1. Triangle ABC has vertices $A(-2, 8)$, $B(-1, 1)$, and $C(-4, 3)$. Find the coordinates of the vertices of $\triangle A'B'C'$ after a rotation of 90° counterclockwise about the origin.



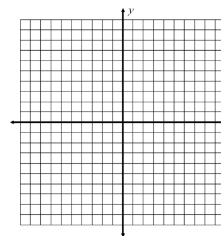
2. Triangle DEF has vertices $D(-7, 1)$, $E(-6, 5)$, and $F(-1, 4)$. Find the coordinates of the vertices of $\triangle D'E'F'$ after a rotation of 270° counterclockwise about the origin.



3. Triangle GHI has vertices $G(0, 0)$, $H(-8, 5)$, and $I(-5, 7)$. Find the coordinates of the vertices of $\triangle G'H'I'$ after a rotation of 180° counterclockwise about the origin.



4. Triangle PQR has vertices $P(8, -2)$, $Q(2, -7)$, and $R(1, -4)$. Find the coordinates of the vertices of $\triangle P'Q'R'$ after a rotation of 540° counterclockwise about the origin.



5. Triangle ABC has vertices $A(-8,1)$, $B(-3,5)$, and $C(-1,2)$. Find the coordinates of the vertices of $\triangle A'B'C'$ after **reflection** across the point $(0,0)$.

The same as
 $R_{180}(x,y) \rightarrow (-x,-y)$

6. Triangle DEF has vertices $D(3,-1)$, $E(5,-7)$, and $F(3,-1)$. Find the coordinates of the vertices of $\triangle D'E'F'$ after rotation of 90° **clockwise** about the origin.

$R_{90}(x,y) \rightarrow (y,-x)$

$D(3,-1) \rightarrow D'(-1,-3)$
 $E(5,-7) \rightarrow E'(-7,-5)$

7. Fill in the missing information.

$R_{90}(4,7) = (\quad , \quad)$	$R_{90}(-2,-8) = (\quad , \quad)$	$R_{90}(9,-5) = (\quad , \quad)$
$R_{180}(4,7) = (\quad , \quad)$	$R_{180}(-2,-8) = (\quad , \quad)$	$R_{180}(9,-5) = (\quad , \quad)$
$R_{270}(4,7) = (\quad , \quad)$	$R_{270}(-2,-8) = (\quad , \quad)$	$R_{270}(9,-5) = (\quad , \quad)$